

Khyathi Vardhan Inturi

Sr. Full Stack Java Developer

<mailto:khyathivardhaninturi@gmail.com> | (513) 273-7943 | [linkedin:inturikhyathivardhan](https://www.linkedin.com/in/inturikhyathivardhan)

PROFESSIONAL SUMMARY

Senior Java Full Stack Developer with 10+ years of experience in designing and developing enterprise applications using Core Java (Java 8+), implementing object-oriented programming principles, multithreading, collections, and performance optimization techniques for high-availability systems. Background in supply chain, logistics, and financial services enterprise environments.

- Extensive experience in building backend systems using Spring, Spring Boot, Spring MVC, Hibernate, and JPA, developing RESTful APIs, implementing Microservices Architecture, and managing dependency injection and transaction management for scalable distributed applications.
- Strong expertise in developing responsive front-end applications using Angular and React along with HTML5, CSS3, JavaScript, and TypeScript, integrating UI components with backend REST services and ensuring cross-browser compatibility and performance optimization.
- Hands-on experience in database design, development, and optimization using Oracle, MySQL, and PostgreSQL, writing complex SQL queries, stored procedures, indexing strategies, and ensuring data integrity.
- Proficient in build and dependency management tools such as Maven and Gradle, managing multi-module projects, versioning, and automated build processes integrated with CI/CD pipelines.
- Strong experience with version control systems using Git, managing branching strategies, code merges, pull requests, and maintaining code quality through peer reviews.
- Experienced in deploying and managing applications on cloud platforms including AWS, Azure, and GCP, configuring cloud services, handling application scaling, and optimizing cloud resource utilization.
- Hands-on experience with containerization and orchestration technologies including Docker and Kubernetes, building container images, managing pods and deployments, and implementing scalable container-based solutions.
- Experience working with messaging systems such as Kafka and RabbitMQ, implementing asynchronous communication, event-driven architecture, and ensuring reliable message processing across microservices.
- Strong understanding of implementing CI/CD pipelines for automated build, testing, and deployment processes, enabling continuous integration and faster release cycles in Agile environments.
- Experience in implementing application security using Spring Security and industry-standard authentication and authorization mechanisms including OAuth2, JWT, and role-based access control, ensuring secure API access.
- Proven ability to lead technical discussions, collaborate with cross-functional teams in Agile/Scrum methodologies, mentor junior developers, and deliver end-to-end solutions aligned with business and technical requirements.
- Strong experience in designing and implementing Microservices Architecture using Spring Boot, enabling service discovery, centralized configuration, API Gateway integration, and inter-service communication.
- Hands-on experience in developing and consuming RESTful APIs with proper request/response handling, exception management, input validation, and API documentation to support seamless integration.
- Extensive experience in implementing data persistence and ORM frameworks using Hibernate and JPA, managing entity relationships, caching strategies, lazy/eager loading, and optimizing transactional operations across Oracle, MySQL, and PostgreSQL databases.
- Practical experience in configuring and managing Docker containers and deploying applications to Kubernetes clusters, handling rolling deployments, replica sets, service exposure, and environment-based configuration management across AWS, Azure, and GCP.
- Experience in setting up and maintaining automated build and deployment workflows using Maven/Gradle integrated with CI/CD pipelines, enabling automated testing, artifact generation, and deployment to cloud environments.
- Strong working knowledge of event-driven systems using Kafka and RabbitMQ, implementing producers and consumers, message serialization, and queue/topic configuration.
- Experience in implementing secure authentication and authorization mechanisms using Spring Security, JWT tokens, and role-based access control to protect REST APIs and ensure secure communication.
- Proven ability to collaborate effectively in Agile/Scrum methodologies, participating in sprint planning, daily stand-ups, code reviews using Git, and delivering production-ready features with performance optimization and cloud deployment support across AWS environments.

TECHNICAL SKILLS

Programming Languages: Java 8 / 11 / 17 / 21, Kotlin, Scala, Python 3, JavaScript (ES6+), TypeScript, SQL (PostgreSQL, MySQL, Oracle), HTML5, CSS3, AJAX, JSON

Java / J2EE Technologies: Servlets, JSP, JSTL, JDBC, RMI, EJB, Multithreading, Serialization, OOP, Java

Frameworks & Libraries: Spring Boot 2 & 3 (4.x familiarity), Spring MVC, Spring Security, Spring Data JPA, Spring Cloud (Netflix Eureka, Gateway), Hibernate ORM / JPA, AngularJS, Angular 5/8/16/17, Angular Material, React.js (Hooks, Redux, React Router), Node.js, Express.js, Apollo Server, Material-UI, Bootstrap 3/4, Tailwind CSS, RxJS

DevOps / CI-CD: Azure DevOps, GitHub Actions, Helm Charts, Service Mesh (Istio, Linkerd), Jenkins, GitLab CI/CD, CircleCI, ArgoCD, Terraform, Ansible, Docker, Kubernetes, Consul, TeamCity, Bamboo, Travis CI, Spinnaker, Octopus Deploy, AWS CodePipeline, Tekton, Harness, Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana)

API Management: Kong, Apigee, Azure API Management, Dell Boomi, IBM iSeries/AS400, JT400/JTOpen, DB2 for i, MQ Series Security: OWASP, SAST, DAST, Zero Trust

Web Technologies: HTML5, CSS3, JavaScript, TypeScript, AJAX, JSON, XML, JSP, React.js, Angular, Node.js, Express.js, Webpack, Databases & Storage: PostgreSQL, MySQL, Oracle 11g/12c, MongoDB, Redis, DynamoDB, Cassandra, MS SQL Server, Amazon Aurora, Elasticsearch, Redshift, Google Cloud SQL, Amazon RDS, Azure Cosmos DB, Vector Databases

Web Services & APIs: RESTful APIs, SOAP, GraphQL, gRPC, Apollo Server, JAX-WS, JAX-RS, Swagger/OpenAPI, API Gateway, Postman, Apache Kafka, WebSocket, Microservices, OAuth 2.0, JWT, Kong, Apigee, NGINX, MuleSoft, Spring Boot, Apache CXF, RabbitMQ

Application & Web Servers: Oracle WebLogic Server (Administration & Deployment), Apache Tomcat, IBM WebSphere, JBoss/WildFly, Jetty, Nginx, Red Hat JBoss EAP, Netty, Spring Boot (Embedded Server), Jakarta EE

Messaging & Streaming: Kafka, Kafka Streams, RabbitMQ, IBM MQ, Azure Event Hub, Confluent Avro, Apache Pulsar, Apache ActiveMQ, NATS, AWS SQS, AWS SNS, AWS Kinesis, Google Cloud Pub/Sub, Azure Service Bus, Redis Streams, Apache Flink, MQTT, AMQP, Apache Camel, Spring Cloud Stream, JMS (Java Message Service)

Cloud & DevOps: AWS (Lambda, S3, Glue, Athena, EMR, CloudFormation, API Gateway, RDS, ECR, ECS, CDK), Azure (Functions, AKS, Key Vault, Event Hub), GCP (BigQuery, Stackdriver, Load Balancer, Cloudflare), Docker, Kubernetes, Terraform, Jenkins, GitLab CI/CD, GitHub Actions, OpenShift, CI/CD pipelines

Testing & QA Tools: JUnit 5, Mockito, Selenium WebDriver, Cypress, Jest, React Testing Library, Cucumber, Postman, JMeter, SOAPUI

Version Control: Git, GitHub, GitLab, Bitbucket, SVN, Azure Repos, AWS CodeCommit

Build & Dependency Management: Maven, Gradle, Ant, npm, Yarn, pip, sbt (Scala Build Tool), Bazel, Nexus Repository

Security & Authentication: OAuth 2.0, JWT, Spring Security, HTTPS, TLS, SAML 2.0, OpenID Connect (OIDC), SSO, Keycloak, Okta, Auth0, AWS IAM, Azure AD, LDAP, MFA/2FA, OWASP, HashiCorp Vault, Firebase Authentication

Monitoring & Performance: Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), AWS CloudWatch, GCP Stackdriver, JVisualVM, Log4j

Methodologies: Agile (Scrum, SAFe), Waterfall, TDD, BDD, CI/CD, Kanban, DevOps, Domain-Driven Design (DDD), Trunk-Based Development, Continuous Delivery, Continuous Deployment

Platforms & OS: Windows, Linux, UNIX

PROFESSIONAL EXPERIENCE

Amazon

Sr. Full Stack Java Developer

Seattle, WA

March 2025 – Present

- Designed and developed scalable backend services using Core Java and Java (17, 21) with multithreading, Streams API, and functional programming for high-performance enterprise applications.
- Built production-grade microservices using Spring, Spring Boot, and Spring MVC, creating RESTful APIs for order processing, inventory management, and payment workflows, ensuring loose coupling and high availability.
- Implemented object-relational mapping and database interactions using Hibernate and JPA, optimizing entity relationships, lazy/eager loading strategies, and transaction management for improved performance with MySQL.
- Implemented robust authentication and authorization mechanisms using Spring Security, configuring role-based access control (RBAC), JWT token validation, and method-level security to protect sensitive business endpoints.
- Developed Angular and React components using TypeScript, implementing reactive forms, route guards, NgRx state management, Redux, and REST API integrations for internal workflow automation and seller-facing operational tools.
- Led front-end architecture decisions for seller-facing React applications, establishing component design standards, folder structure conventions, shared vs. feature boundaries, and enforcing coding guidelines through structured code reviews and pull request workflows to improve codebase consistency and long-term maintainability.
- Drove incremental migration of legacy class-based React components to functional components with Hooks, managing coexistence of old and new UI patterns without big-bang rewrites, and optimizing bundle size and rendering performance across seller-facing enterprise dashboards.
- Developed lightweight Node.js and Express services to support server-side rendering and API proxying for React-based front-end applications, reducing client-side latency and enabling server-driven data pre-fetching to optimize load performance across high-traffic operational workflows.
- Designed event-driven communication between microservices using Apache Kafka, producing and consuming messages for real-time order updates, asynchronous processing, and decoupled system integration.
- Built and maintained CI/CD pipelines using Jenkins, automating code build, unit testing, static code analysis, and container image generation to ensure reliable and repeatable deployments.
- Containerized applications using Docker and deployed to AWS cloud infrastructure, leveraging AWS services such as EC2, S3, RDS (MySQL), and EKS for scalable and fault-tolerant deployments.
- Implemented GitOps-based deployment strategies using ArgoCD, enabling automated Kubernetes deployments, version-controlled infrastructure, and environment synchronization.

- Utilized New Relic for application performance monitoring (APM), identifying memory bottlenecks, slow database queries, and latency issues across distributed microservices.
- Applied performance tuning techniques at JVM, database, and API levels, optimizing thread pools, connection pooling, and query indexing to improve response times and throughput.
- Built and maintained AWS data pipelines using Glue, S3, and Athena for large-scale data ingestion and transformation; developed Python 3 scripts for ETL automation and data validation, integrating with Lambda and Glue workflows to support backend data processing at scale.
- Integrated Generative AI capabilities into backend services using OpenAI and AWS Bedrock APIs building RAG pipelines with LangChain for intelligent document retrieval and context-aware responses, implementing prompt engineering patterns, and leveraging GitHub Copilot to accelerate code generation, testing, and documentation across the engineering team.
- Worked with Palantir Foundry and Palantir AIP on AWS, building backend data connectors and TypeScript-based dashboards to surface AI-driven analytics and operational insights for internal enterprise workflows.

Environment: Core Java, Java (17,21), Spring, Spring Boot, Spring MVC, Hibernate, JPA, Microservices Architecture, RESTful APIs, Angular, React, Node.js, Express.js, TypeScript, MySQL, Aurora MySQL, Docker, Kubernetes (EKS), AWS, AWS Glue, Amazon Athena, AWS Lambda, S3, IAM, Jenkins, Git, Maven, Gradle, Apache Kafka, JUnit, ArgoCD, Prometheus, Grafana, New Relic, Spring Security, Python 3, OpenAI, LangChain, AWS Bedrock, GitHub Copilot, Palantir Foundry, Palantir AIP

Northern Trust

Sr. Full Stack Java Developer

Chicago, IL

May 2022 – February 2025

- Developed enterprise-grade backend applications using Core Java and Java 17, leveraging Streams API, multithreading, and concurrency utilities to build secure and high-performance financial transaction processing systems.
- Designed and implemented scalable microservices using Spring, Spring Boot, and Spring MVC, exposing standardized RESTful APIs for portfolio management, client reporting, and transaction workflows within a distributed Microservices Architecture.
- Integrated persistence layers using Hibernate and JPA, configuring entity mappings, caching strategies, and optimized JPQL/native queries to ensure efficient data access with MySQL databases.
- Leveraged Oracle 12c as the primary relational database for core banking and portfolio management services, writing advanced SQL queries, stored procedures, and optimizing execution plans for high-volume financial transaction processing via JDBC and optimized connection pooling.
- Built responsive and dynamic front-end components using Angular and TypeScript, integrating REST APIs, implementing reactive forms, HTTP interceptors for token handling, and dashboard visualizations for financial reporting modules.
- Built Node.js utility services to handle data formatting, report generation, and API response transformation between Angular front-end dashboards and Spring Boot backend services, reducing client-side processing overhead for financial reporting modules.
- Strengthened application security by configuring advanced OAuth2 flows within Spring Security context, securing Angular-based UI integrations, and enforcing token validation and role-based access control while maintaining automated quality checks through JUnit, Jenkins, Git, Maven, and Gradle pipelines.
- Deployed containerized microservices to Microsoft Azure, leveraging Azure Kubernetes Service (AKS), Azure App Services, and Azure SQL integrations to ensure scalable and secure cloud-based infrastructure.
- Implemented GitOps-based continuous delivery practices using Flux, synchronizing Kubernetes clusters with Git repositories for automated and version-controlled production deployments.
- Integrated AppDynamics for application performance monitoring, identifying transaction bottlenecks, memory leaks, and slow database queries within distributed financial systems.
- Applied performance optimization techniques at application and database levels, profiling JVM heap, GC tuning, optimizing MySQL indexing strategies, and improving API response times in high-volume financial operations.
- Designed and maintained data pipelines using Apache NiFi for high-volume financial data ingestion, transformation, and persistence into MySQL, integrating with Core Java and Spring MVC backend services to support near real-time processing across banking systems.
- Built enterprise integration workflows using Dell Boomi to connect financial systems across cloud and on-premises environments on Azure, implementing data mappings, transformation logic, error handling, and monitoring to meet compliance and governance standards.
- Integrated modern Java/Spring Boot services with legacy IBM iSeries (AS/400) banking systems using JT400 and JDBC connectivity consuming REST and SOAP web services, MQ-based messaging, and JDBC/JT400 connectors to bridge legacy DB2 for i data with modern Java microservices; developed batch processing and file exchange workflows using IFS (Integrated File System); maintained SLA-driven production support aligned with ITSM change management on Azure.
- Developed Python 3 automation scripts for data extraction, transformation, and reporting workflows, integrating with Azure services and internal financial data pipelines to accelerate reporting cycles and reduce manual processing overhead.

Environment: Core Java, Java 17, Spring, Spring Boot, Spring MVC, Hibernate, JPA, Microservices Architecture, RESTful APIs, Angular, Node.js, TypeScript, MySQL, Oracle 12c, Docker, Azure, AKS, Jenkins, Git, Maven, Gradle, Apache NiFi, Dell Boomi, IBM iSeries, JT400, IBM MQ, DB2 for i, JUnit, Flux, Prometheus, Grafana, AppDynamics, OAuth2, Python 3, Spring Security

HCA Healthcare

Full Stack Java Developer

Nashville, TN

July 2019 – April 2022

- Developed enterprise healthcare applications using Core Java and Java 11, applying multithreading, Streams API, and concurrency utilities to process patient records, claims data, and clinical workflows in a distributed Microservices Architecture.
- Designed and implemented backend services using Spring, Spring Boot, and Spring MVC, exposing secure and standardized RESTful APIs for patient management, appointment scheduling, and billing systems.
- Built robust persistence layers using Hibernate and JPA, mapping complex healthcare entities and optimizing query performance with proper indexing and transaction management in MySQL databases.
- Performed and optimized enterprise data models using Oracle databases, writing complex SQL queries, stored procedures, and performance tuning to support Life Sciences and clinical data workflows including patient records, clinical trial data management, and regulatory reporting systems aligned with healthcare compliance standards.
- Developed reusable business components using Jakarta EE specifications, integrating servlets and enterprise APIs with Spring-based microservices to support legacy and modern healthcare modules.
- Implemented secure authentication and authorization mechanisms using Keycloak, configuring role-based access control (RBAC), token-based authentication, and single sign-on (SSO) integration for internal healthcare portals.
- Developed responsive and interactive user interfaces using React, integrating REST endpoints, managing application state, and implementing secure API communication aligned with backend authentication mechanisms.
- Implemented robust exception handling and validation frameworks within Spring Boot and Spring MVC services, standardizing error responses across RESTful APIs and ensuring secure communication between React front-end modules and backend systems protected by Keycloak.
- Designed and maintained event-driven data ingestion pipelines using Apache Flume, capturing and transporting healthcare system logs and transactional data to centralized storage and processing systems.
- Developed Java-based integration services connecting modern Spring Boot microservices with legacy IBM iSeries (AS/400) hospital billing, pharmacy, and clinical data systems consuming SOAP and REST APIs, integrating DB2 for i data via JDBC/JT400 connectors, and processing batch patient data and claims through IFS-based file exchange workflows; supported SLA-driven production operations and ITSM change management processes on GCP.
- Administered and deployed Spring Boot applications to Oracle WebLogic Server, managing server configuration, deployment descriptors, datasource setup, and application lifecycle management across development, QA, and production environments.
- Deployed containerized microservices to Google Cloud Platform (GCP) using services such as GKE, Compute Engine, and Cloud SQL, ensuring scalable, secure, and highly available cloud infrastructure.
- Built CI/CD pipelines using Jenkins, automating compilation, unit testing, artifact packaging, and deployment processes to streamline release cycles.
- Implemented continuous delivery strategies using Spinnaker, orchestrating blue-green and rolling deployments across GCP clusters with rollback capabilities, ensuring minimal downtime and seamless integration of Jakarta EE, Spring, and containerized microservices environments.
- Configured performance monitoring and distributed tracing using Dynatrace, identifying bottlenecks in microservices, database calls, and REST endpoints to improve overall system performance.
- Applied performance tuning techniques across Spring Boot services, optimizing JVM configurations, connection pooling, and SQL query execution plans strategies to enhance application throughput.
- Participated in Agile development processes, collaborating with cross-functional healthcare teams, conducting code reviews, resolving production issues, and ensuring secure and reliable enterprise application delivery.

Environment: Core Java, Java 11, Spring, Spring Boot, Spring MVC, Hibernate, JPA, Microservices Architecture, RESTful APIs, React, MySQL, Oracle, Docker, GCP, GKE, Jenkins, Git, Maven, Gradle, Apache Flume, Jakarta EE, Spinnaker, Dynatrace, Keycloak, Oracle WebLogic Server, IBM iSeries, JT400, DB2 for i, IFS, Life Sciences

Southwest Airlines

Full Stack Java Developer

Dallas, TX

December 2016 – June 2019

- Developed backend services using Core Java and Java 8, implementing multithreading, Streams API, and collections framework to support booking, ticketing, and flight scheduling systems within a scalable Microservices Architecture.
- Designed and built REST-based microservices using Spring, Spring Boot, and Spring MVC, exposing secure and reliable RESTful APIs for reservations, fare calculations, and passenger information management.
- Implemented persistence layers using Hibernate and JPA, mapping complex airline domain entities and optimizing queries for high-volume transactions in MySQL databases.

- Developed dynamic and responsive user interfaces using React, integrating backend REST services, managing component state, and ensuring seamless user interactions for booking and check-in modules.
- Utilized Git for version control, implementing feature branching, performing peer code reviews, and maintaining codebase stability across development and production releases.
- Built and maintained CI/CD pipelines using Jenkins, automating compilation, unit testing, packaging, and deployment workflows to ensure continuous integration and reliable release cycles.
- Deployed applications to Google Cloud Platform (GCP), leveraging Compute Engine and Cloud SQL (MySQL) for scalable infrastructure, secure networking, and high availability of airline service applications.
- Optimized application performance by tuning Spring Boot configurations, managing connection pools, refining Hibernate/JPA queries, and reducing API response times under peak booking loads.
- Implemented standardized exception handling, logging, and request validation mechanisms within Spring MVC controllers to ensure consistent API responses and maintain data integrity.
- Collaborated with cross-functional teams in Agile development environments, participating in sprint planning, backlog refinement, code reviews, and production support to deliver stable and maintainable airline enterprise applications.

Environment: Core Java, Java 8, Spring, Spring Boot, Spring MVC, Hibernate, JPA, Microservices Architecture, RESTful APIs, React, MySQL, GCP, Jenkins, Git, Maven, Gradle

Mastek
Java Developer

Mumbai, India
May 2015 – October 2016

- Developed enterprise web applications using Core Java and Java 8, implementing object-oriented design principles, multithreading, and collections framework to support scalable and maintainable business modules across service-oriented architectures.
- Designed backend services using Spring, Spring Boot, and Spring MVC, exposing structured and secure RESTful APIs for client-facing and internal system integrations.
- Built and optimized persistence layers using Hibernate and JPA, mapping relational entities, configuring caching strategies, and managing transactions with MySQL databases to ensure reliable data consistency.
- Developed reusable enterprise components using Jakarta EE, integrating servlets and enterprise APIs with Spring-based services to support hybrid architecture requirements.
- Implemented secure authentication and authorization using Spring Security, configuring role-based access control, session management, and form-based login mechanisms to protect enterprise application endpoints.
- Developed responsive front-end interfaces using React, integrating REST endpoints, managing component lifecycle methods, and ensuring secure communication with backend services.
- Designed and maintained data ingestion workflows using Apache Flume, capturing application logs and streaming data into centralized systems for further analysis and monitoring.
- Deployed and managed applications on AWS, utilizing EC2 for compute resources and RDS (MySQL) for managed database services, ensuring scalable and reliable infrastructure.
- Built and maintained CI/CD pipelines using Jenkins, automating build, test, and deployment workflows across AWS environments to ensure consistent and reliable application releases.
- Integrated centralized logging and monitoring using the ELK Stack (Elasticsearch, Logstash, Kibana) to collect, index, and visualize application logs for troubleshooting and operational insights.
- Configured performance monitoring and transaction tracing using Dynatrace, identifying bottlenecks in application layers, database queries, and REST endpoints to improve overall system stability.
- Applied performance optimization techniques within Spring Boot services, tuning JVM configurations, optimizing Hibernate/JPA queries, and refining MySQL indexing strategies to enhance response time and throughput.
- Participated in requirement analysis, design discussions, code reviews, and production support activities, ensuring delivery of secure, maintainable, and scalable full-stack enterprise solutions.

Environment: Core Java, Java 8, Spring, Spring Boot, Spring MVC, Hibernate, JPA, Microservices Architecture, RESTful APIs, React, MySQL, AWS, Apache Flume, Jakarta EE, Jenkins, ELK Stack, Dynatrace, Spring Security

Education:

Vellore Institute of Technology (2011-2015)
Bachelor of Technology in Computer Science